HEALTHCARE AI · HACKATHON SUBMISSION

GlucoTwin.

An AI patient digital twin that forecasts glucose 2 hours ahead and flags the risk of a low or a high, with reasons.

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RESEARCH PROTOTYPE · SYNTHETIC DATA

TEAM & INSTITUTION

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PROBLEM STATEMENT

Dangerous lows and highs are often visible in the data before they happen, but that data is split across systems.

THE HEALTH RECORD (EHR)

Who the patient is

Diabetes type, HbA1c, medications, history of hypoglycaemia.

CGM + WEARABLE

What is happening now

Glucose trend, heart rate, activity, sleep, meals and insulin.

TODAY'S ALERTS

React, don't anticipate

Clinicians rarely see both sources together, and most CGM alerts react to the current value instead of the next two hours.

OUR QUESTION

Can a patient digital twin that fuses static EHR context with live sensor streams forecast glucose 2 hours ahead, flag threshold crossings, and show whether each data source actually helps?

HEALTHCARE USE CASE

Decision support for diabetes care teams monitoring patients on CGM

WHO AND HOW

- **Users:** endocrinologists, diabetes nurses and remote-monitoring teams caring for people with type 1 or type 2 diabetes.
- **Workflow:** sensor data arrives every 5 min → the twin updates → forecast, risk and reasons appear on one screen → the clinician reviews.
- **Value:** earlier warning of lows and highs, EHR and sensor context in one view, uncertainty always shown, explanations to check before acting.
- **Safety:** informational only; it never gives treatment or insulin-dosing advice.

EXAMPLE · HELD-OUT PATIENT SYN-023 · T1D · 22 Y

140	102	21%
mg/dL now, rising	forecast in 2 h	hypo risk · elevated

Glucose is still rising after a meal, yet the twin expects it to fall. Top drivers: **insulin logged in the last 2 h** and **insulin on board**. A trend-only alarm would miss this.

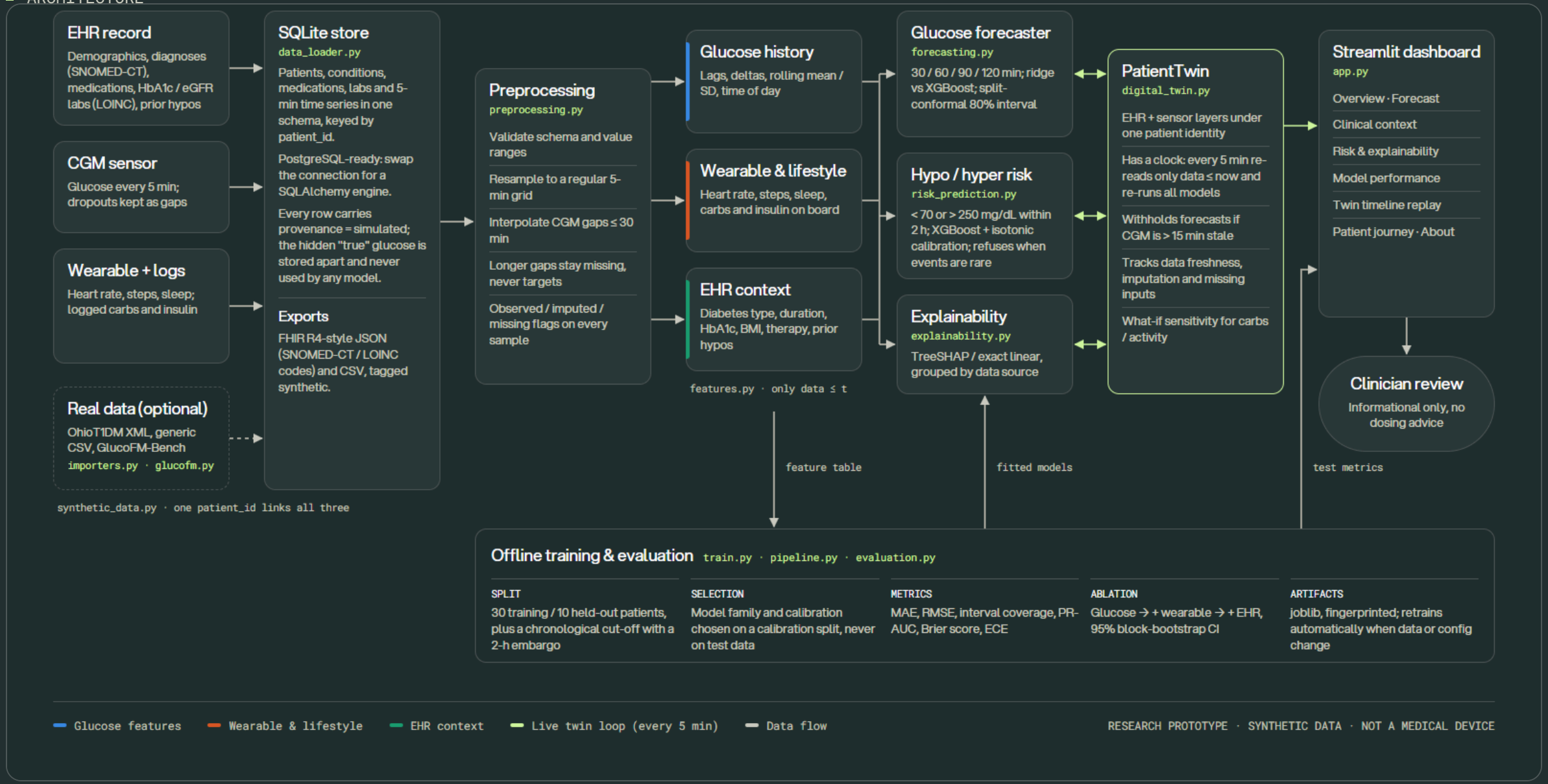
OUR SOLUTION

A stateful virtual patient that updates every 5 minutes

GlucoTwin combines each patient's EHR record with their CGM and wearable stream, forecasts the next 2 hours with uncertainty, estimates calibrated low / high risk, and explains every prediction in a doctor-facing dashboard.

DIFFERENTIATOR	WHAT WE BUILT
A real twin, not a static chart	PatientTwin re-reads only data up to its clock and re-runs every model at each update; a replay shows it evolve live
Measured value of data fusion	Three-way ablation (glucose → + wearable → + EHR) with block-bootstrap confidence intervals
Scientific hygiene	Held-out patients and a chronological cut-off with embargo; selection and calibration never touch test data
Honest uncertainty	Conformal prediction intervals; risk models refuse to output probabilities when events are too rare
Safety by design	Withholds forecasts on stale CGM; labels observed / imputed / simulated data; no dosing advice
Interoperability-ready	SQLite → PostgreSQL path; FHIR R4-style export with SNOMED-CT / LOINC codes

ARCHITECTURE



FULL-SIZE DIAGRAM: DOCS/GLUCOTWIN_ARCHITECTURE.PDF

TECHNICAL STACK

Pure Python, runs offline on a laptop

LANGUAGE & DATA

Python 3.10+ (tested on 3.14)

pandas, NumPy, SciPy

SQLite, PostgreSQL-ready

FHIR R4-style JSON, CSV

MACHINE LEARNING

XGBoost (gradient boosting)

scikit-learn: ridge, isotonic calibration

Split-conformal intervals

TreeSHAP explanations

APP & QUALITY

Streamlit dashboard

Plotly charts

pytest: causality and leakage tests

joblib artifacts, fingerprinted

DATA SOURCES

Synthetic cohort generator; importers for OhioT1DM XML and generic CSV; GlucoFM-Bench via Hugging Face datasets.

SPEED

Data generation ≈ 6 s · full training ≈ 30 s · one twin update ≈ 0.3 s.

DEPENDENCIES

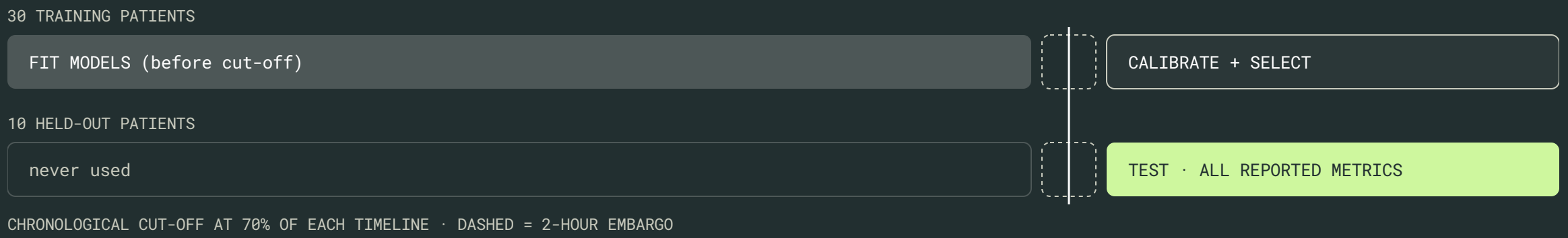
No paid APIs, no cloud. XGBoost is optional, with an automatic fallback to scikit-learn.

Simple first; add complexity only if it helps

COMPONENT	CHOICE	WHY
Forecast target	Observed CGM 30 / 60 / 90 / 120 min ahead; separate direct model per horizon, predicting the change	No error compounding; imputed values are never targets
Model families	Ridge regression vs XGBoost, chosen by calibration-set MAE	Never tuned on test data
Feature sets	① glucose only ② + wearable & lifestyle ③ + EHR (full twin), vs persistence	Measures what each source adds
Uncertainty	Split-conformal 80% prediction intervals	Distribution-free; coverage reported
Risk	New crossing of < 70 or > 250 mg/dL in 2 h; XGBoost + isotonic calibration	Calibrated probabilities, F1-max alert threshold
Refusal	< 30 training or < 10 calibration events → no probability shown	No invented probabilities
Explanations	TreeSHAP (XGBoost pred_contribs); exact linear terms for ridge	Labelled as associations, not causes

EVALUATION DESIGN

Every reported number comes from patients and times the models never saw



01 · PATIENTS

Test patients are never used for training, model selection or calibration.

02 · TIME

Chronological cut-off with a 2-hour embargo so no target leaks across it.

03 · PREPROCESSING

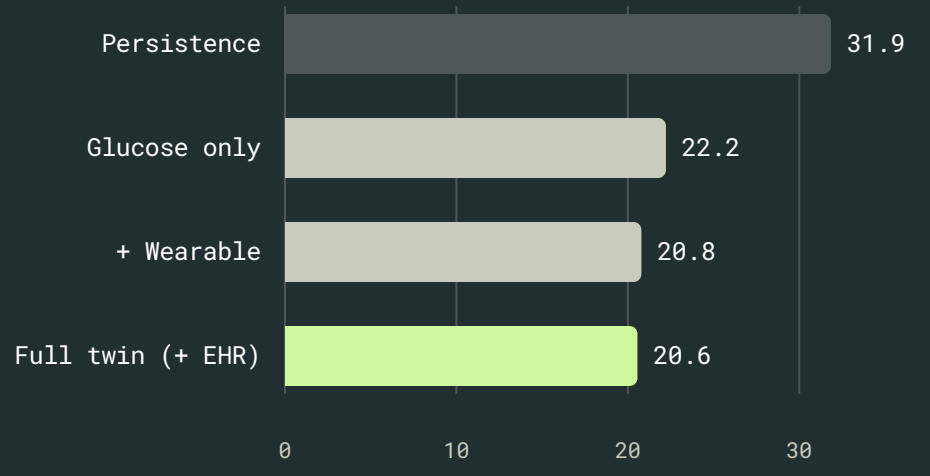
Imputers, scalers and calibrators are fitted on training data only.

04 · TESTS

Automated tests check causal features, no patient overlap and the embargo.

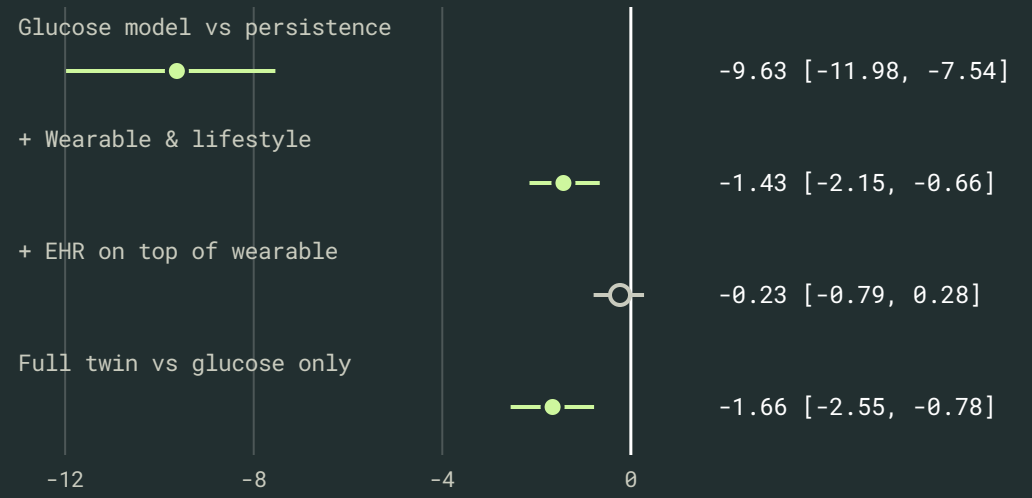
The full twin cuts 2-hour forecast error by 35% versus persistence

2-HOUR MAE, HELD-OUT PATIENTS (MG/DL, LOWER IS BETTER)



n = 11,625 test points · selected model per configuration: XGBoost

ABLATION: Δ MAE WITH 95% BLOCK-BOOTSTRAP CI



Left of 0 = lower error. EHR on top of wearable data: CI includes 0, so no clear gain.

Calibrated risk, reported with its limits

EVENT (NEXT 2 H)	CONFIGURATION	PR-AUC	RECALL	BRIER
Low < 70 · base 1.1%	Glucose only	0.46	0.57	0.007
	+ Wearable	0.38	0.64	0.008
	Full twin	0.31	0.42	0.009
High > 250 · base 7.6%	Glucose only	0.50	0.52	0.050
	+ Wearable	0.53	0.50	0.049
	Full twin	0.51	0.46	0.051

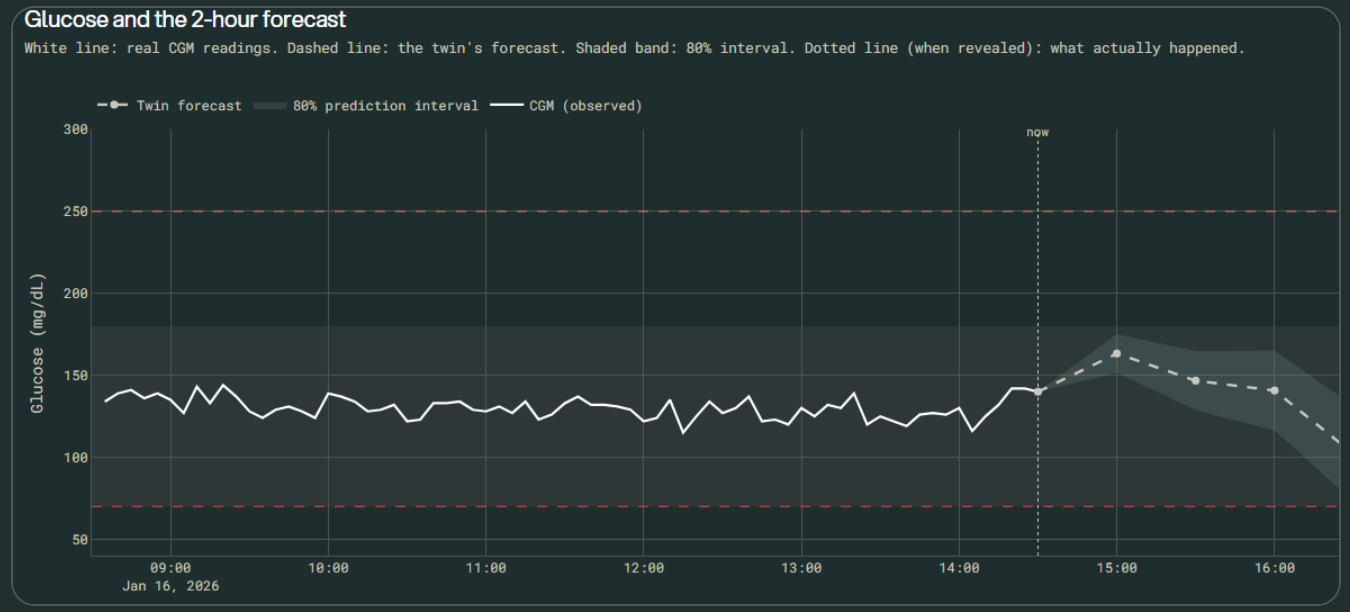
PR-AUC of a random model equals the base rate (0.011 for lows, 0.076 for highs).

HOW TO READ THIS HONESTLY

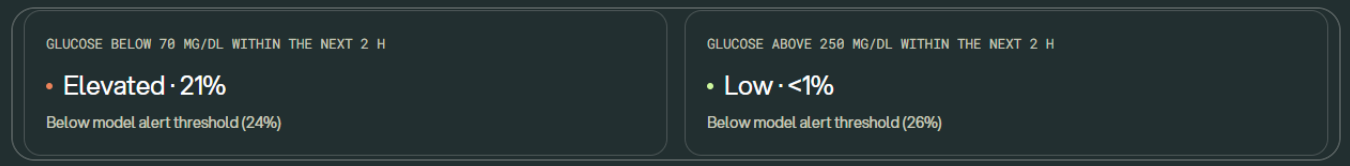
- Wearable signals clearly help the forecast; EHR context adds little once wearable data is present.
- Lows are rare (125 test windows); adding static EHR features made the hypo model worse, consistent with overfitting on 30 training patients.
- Interval coverage is 0.76–0.81 against a nominal 0.80.
- Synthetic results show the pipeline works; they are not clinical evidence.

LIVE DEMO WALKTHROUGH

Seeing a low before it happens



FORECAST · DASHED PATH WITH 80% BAND FALLS TOWARDS 70 MG/DL



RISK · HYPO RISK ELEVATED (21%) WHILE GLUCOSE IS STILL 140 AND RISING

01 **Load** held-out patient SYN-023; the SIMULATED badge is always visible.

02 **Clinical context**: diagnoses, insulin therapy, HbA1c history and 24 h of wearable data.

03 **Next alert** jumps to just before the model's next low-glucose alert.

04 **Forecast** shows the 2-hour trajectory and its 80% interval.

05 **Risk & XAI** explains why: insulin logged and still on board.

06 **Timeline ▶ Play** streams new data; past forecasts are plotted against what happened.

FULL DEMO VIDEO: LINK IN README

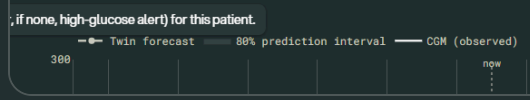
Overview

Current state of the selected patient's digital twin: latest glucose, the 2-hour forecast, risk and data quality at a glance.

GLUCOSE NOW	FORECAST +2 H	TIME IN RANGE	VARIABILITY	RISK (2 H)
140 →	102	94%	12%	Elevated
mg/dL · +0.5 mg/dL/min	mg/dL · interval 73-132	below 0.0% · above 6%	CV · mean 147 mg/dL · GMI 6.8%	hypo 21% · hyper 0%
Latest CGM reading and its trend per minute.	Predicted glucose in 2 hours, with its 80% interval.	Share of the last 24 h between 70 and 180 mg/dL.	Coefficient of variation; above 36% means unstable glucose.	Higher of the low and high risk categories for the next 2 hours.

Digital-twin snapshot

White: real CGM. Dashed: the twin's forecast. Band: 80% interval.



Synthetic patient profile

Static health-record facts used by the twin.

FIELD	VALUE
Diabetes	T1D, 2 y since diagnosis

OVERVIEW · STATE AT A GLANCE

Clinical Context

The health-record and wearable context the twin uses: diagnoses, medications, labs, heart rate, steps, sleep, meals and insulin.

Wearable & logged lifestyle (last 24 h)

Heart rate; steps with sleep marked; logged carbs and insulin. Heart rate and steps are missing for 5% of the period (e.g. watch charging) and are left missing, not set to zero.



Diagnoses

Coded conditions from the health record.

CONDITION	SYSTEM	CODE	ONSET
Diabetes mellitus type 1	SNOMED-CT	46635009	2024-01-06

Medications

Current medicines and their drug class.

MEDICATION	CLASS	STATUS
Rapid-acting insulin via pump (CSII)	insulin	active

Laboratory results

Lab tests recorded before monitoring started.

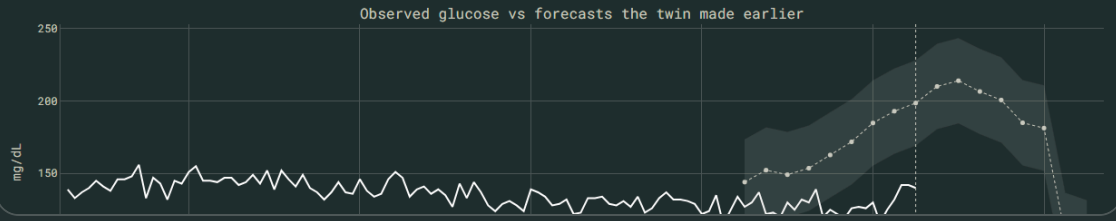
DATE	TEST	VALUE	UNIT

CLINICAL CONTEXT · EHR + WEARABLE

GLUCOSE NOW	FORECAST +2 H	HYPO RISK (2 H)	HYPER RISK (2 H)
140 mg/dL	102 mg/dL	21%	0%
↑ +8	↓ -5	↑ +2%	↓ -0%

Earlier forecasts vs what happened

Top: real glucose and the 2-hour forecasts plotted at the time they refer to. Bottom: risk over the replay. The twin only sees data left of the dotted 'now' line.



TWIN TIMELINE · LIVE REPLAY

Forecasting accuracy (test patients)

Mean absolute error (MAE) and RMSE in mg/dL for each configuration and model at the chosen horizon; lower is better.

Horizon

○ 30 min ○ 60 min ○ 90 min ● 120 min

CONFIGURATION	MODEL	USED BY TWIN	MAE	RMSE	CALIBRATION MAE	80% INTERV
0 - Persistence (last value)	Naive	—	31.85	42.67	—	
1 - Glucose only	Ridge regression	—	24.57	33.4	25.66	
1 - Glucose only	Gradient boosting (XGBoost)	yes	22.22	31.76	22.35	
2 - Glucose + wearable	Ridge regression	—	22.96	30.49	23.82	
2 - Glucose + wearable	Gradient boosting (XGBoost)	yes	20.79	29.23	19.71	
3 - Full digital twin (+ EHR)	Ridge regression	—	21.02	28.6	22.16	
3 - Full digital twin (+ EHR)	Gradient boosting (XGBoost)	yes	20.56	29.04	19.01	

MODEL PERFORMANCE · HELD-OUT PATIENTS

What we guard against, and what remains open

RISK / LIMITATION	MITIGATION IN THIS PROTOTYPE	NEEDED NEXT
Synthetic data encodes our assumptions	Every assumption listed in-app; results framed as a pipeline demonstration	External validation on real CGM + EHR data
Data leakage	Patient + chronological split, 2-h embargo, train-only preprocessing, tests	Prospective validation
Rare hypo events	Minimum-event rules; refuse to show probabilities; report base rates	Larger cohorts, event-level metrics
Missing / stale sensor data	Explicit flags; gaps > 30 min never imputed; forecast withheld if CGM > 15 min old	Device-level quality signals
Over-trust in explanations	Labelled as associations; method named on screen	Clinician user studies
Privacy, bias, regulation	Synthetic only; research positioning; no dosing advice	De-identification, subgroup audits, V&V and regulatory review

What we delivered, and where it goes next

DELIVERED

- End-to-end digital twin: data → features → models → twin → dashboard
- 2-h MAE 20.6 mg/dL vs 31.9 for persistence on unseen patients
- Measured fusion gain: wearable data helps significantly; EHR gain not significant, reported as such
- Calibrated risk with refusal, conformal intervals, TreeSHAP reasons
- Real-data importers and FHIR-style export ready

NEXT

- Validate on real CGM data (OhioT1DM under its data-use agreement, or a clinical partner)
- Subgroup and fairness audits on representative data
- Hybrid mechanistic + ML twin
- Event-level alert metrics: lead time, false alerts per day
- Usability sessions with clinicians; FHIR-native EHR and streaming ingestion

GlucoTwin.

Seeing a low before it happens.

CODE

github.com/DakshKumar05/glucotwin

DEMO VIDEO

Unlisted YouTube link in the README

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